THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. 40 g of salt is added to 160 g of water. The mass percentage of the solution is: 1

(a) 25 per cent **(b)** 40 per cent **(c)** 20 per cent **(d)** 10 per cent

2. A beam of light passes through a mixture and becomes visible. The mixture is a: 1

(a) true solution **(b)** gas **(c)** pure compound **(d)** colloid

3. A student prepares tea with 5 g sugar in 100 mL and another with 10 g sugar in 100 mL. The second tea is: 1

(a) a suspension **(b)** more dilute
(c) saturated **(d)** more concentrated

4. How many grams of sugar are in 250 g of a 12 per cent sugar solution? 1

(a) 30 g **(b)** 25 g **(c)** 12 g **(d)** 3 g

5. Which of the following will show the Tyndall effect? 1

(a) salt solution **(b)** milk **(c)** vinegar **(d)** sugar solution

6. In a fair test on the effect of stirring, which must be kept the same? 1

(a) colour **(b)** stirring
(c) temperature and amount of sugar **(d)** time to dissolve

7. How much water is needed to make 500 g of a 5 per cent salt solution? 1

(a) 450 g **(b)** 475 g **(c)** 500 g **(d)** 25 g

8. Which of these settles on standing? 1

(a) sand in water **(b)** sugar in water **(c)** salt in water **(d)** milk

9. The solubility of salt at 20 degrees Celsius is 36 g per 100 g of water. If 40 g is added to 100 g of water at 20 degrees Celsius, then: 1

(a) 40 g remains **(b)** 36 g remains
(c) 4 g remains undissolved **(d)** all dissolves

10. A solution contains 15 g of solute in 135 g of solvent. The mass percentage is: 1

(a) 10 per cent **(b)** 15 per cent **(c)** 9 per cent **(d)** 11 per cent

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the mass percentage of a solution made by dissolving 18 g of sugar in 82 g of water. 2

12. Why is stirring not changing the solubility of a substance? 2

13. Why does a lemon-water solution not scatter light but milk does? 2

14. How much salt is required to prepare 250 g of 8 per cent solution? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A solution is prepared by dissolving 35 g of salt in 140 g of water. Find the mass of the solution and the mass percentage of salt. 3

16. Describe the steps for obtaining crystals of copper sulphate from its saturated solution. 3

17. Describe three methods for separating components of solutions. 3

OR

How much sugar must be dissolved in 360 g of water to prepare a 10 per cent solution? 3

18. Explain why oil floats on water and does not dissolve in it. Name a solvent for it. 3

OR

Why is distilled water used in car batteries and irons? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe an experiment to find out how the solubility of salt and of sugar changes with temperature. State the variable, procedure, table and conclusion. 5

OR

Explain the changes that occur when a saturated solution of KNO3 at 60 degrees Celsius is cooled to 20 degrees Celsius. 5

20. A 250 g solution has 40 g of solute. Find the percentage. If 50 g of water is evaporated find the new percentage. If 100 g of water is added to the original solution find its percentage. 5

OR

Explain the process of obtaining pure water from sea water in the laboratory and in nature. 5

END OF QUESTION PAPER — Total: 40 Marks

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